Capgemini engineering

TETRABOT
TEAM TRANSPORT BY ROBOTS

TETRABot (TEam TRAnsportation roBOT)

ENGINEERING HORIZONS

A conference for engineers by engineers

THE CHALLENGE: SPECIAL CONDITIONS & REQUIREMENTS WITHIN AIRCRAFT INTEGRATION

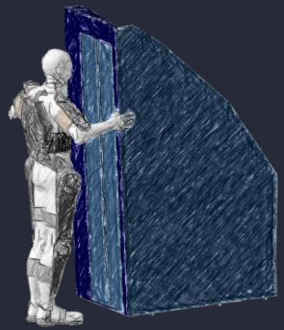
Lack of automation during aircraft integration (especially for monuments)

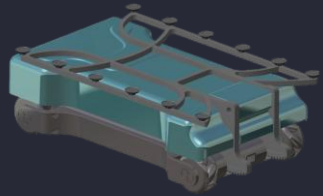
- poor ergonomic conditions
- narrow and poorly accessible workspaces
- sensitive and expensive floor (load limit 25 kg / cm²)
- small batches but high variance
- dynamic and difficult to predict environment

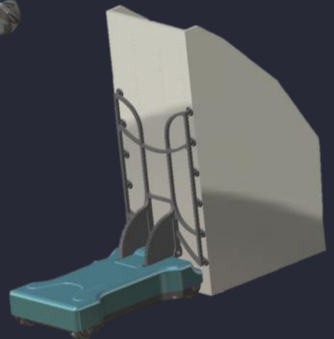
IDEA GENERATION FOR SOLUTIONS TO OVERCOME THE TRANSPORT PROBLEM

Investigation in different directions to find solutions

- Exo skeleton (ergonomically driven)
- AGV (Automated Guided Vehicles)
- Extended AGV

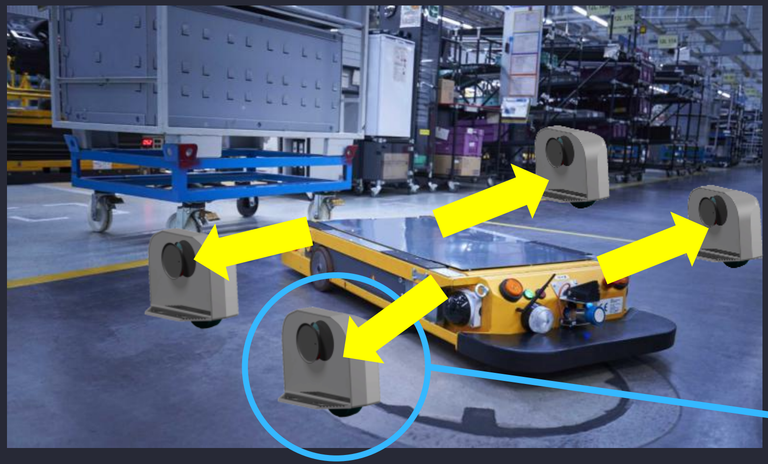


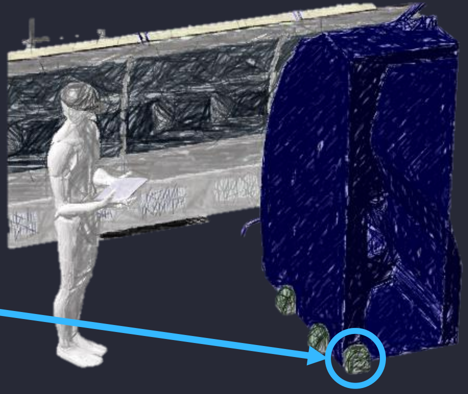




AGV: nice & established solution … **BUT**:

- Too heavy for aircraft cabin
- Not very flexible for different payloads



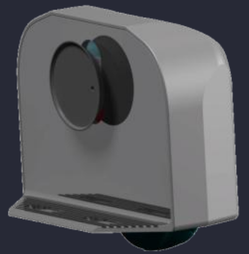


"Make the payload part of the AGV!"

(weight reduction)

EVOLUTION AND DESIGN OPTIMISATION

- one (regular) wheel
- simple docking concept

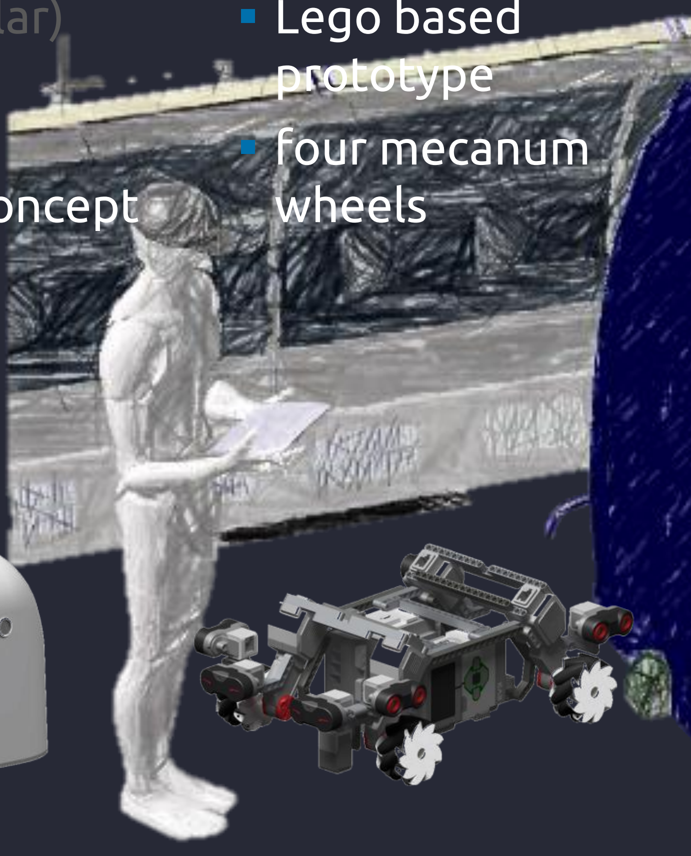


2018

- one (regular) wheel
- improved docking concept

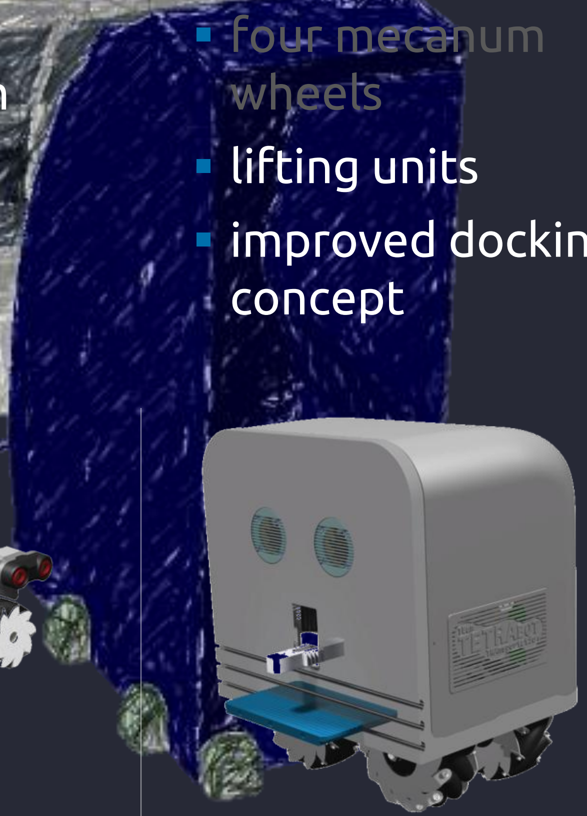
2019

- Lego based prototype
- four mecanum wheels



2020

- first full prototype
- four mecanum wheels
- lifting units
- improved docking concept



2020

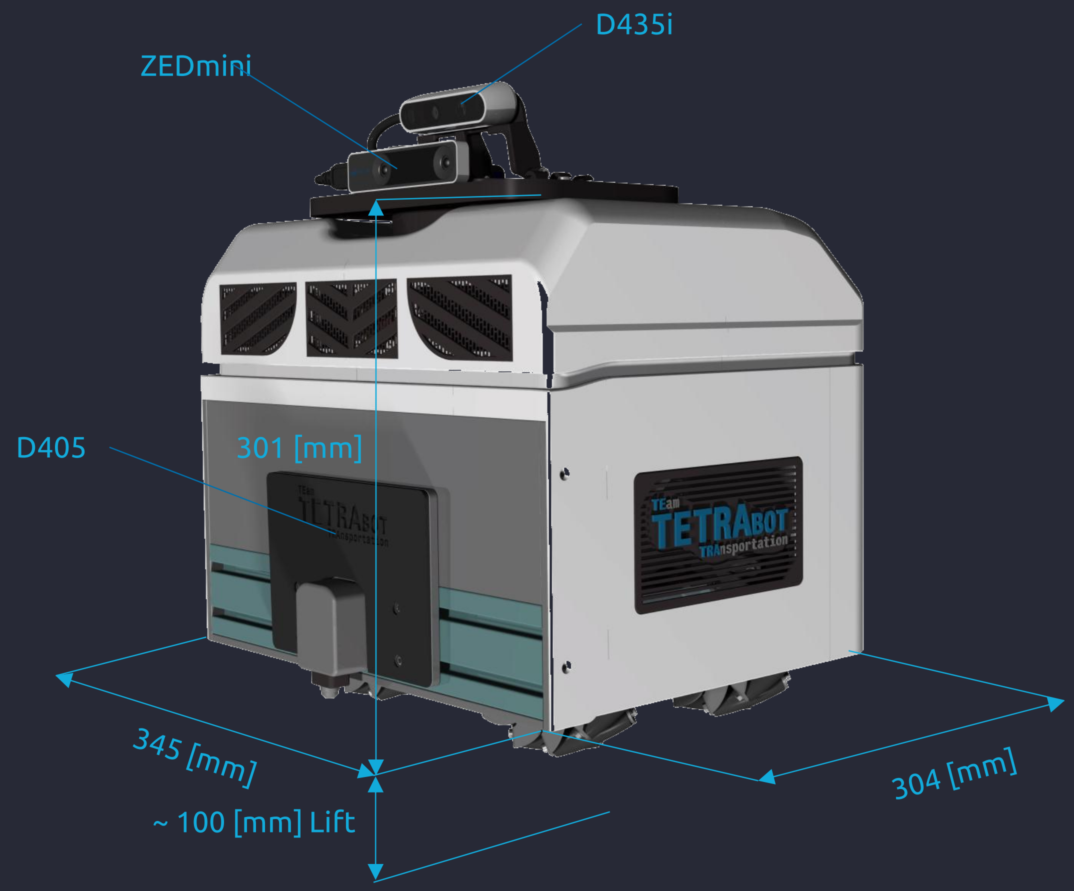
- …
- prototype next generation
- start of sensor integration

2021

TECHNICAL FEATURES

Features

- Own weight: less 20 kg
- Size: 345 x 304 x 301 mm
- Lift load: ~ 80 kg
- Lift height: ~ 100 mm
- Mecanum wheels: 3DOF in the plane
- Max speed: ~ 5 km/h (depending on load and environment)
- Computer: Jetson Nano
- Operating System: Ubuntu, ROS1
- Communication: WLAN
- Sensors:
  - Aligned depth & RGB camera, Intel Realsense D435i
  - 6 DOF tracking sensor, ZEDmini
  - Docking Camera Intel D405
  - OpenCV
- max. battery runtime: ~ 2h operational



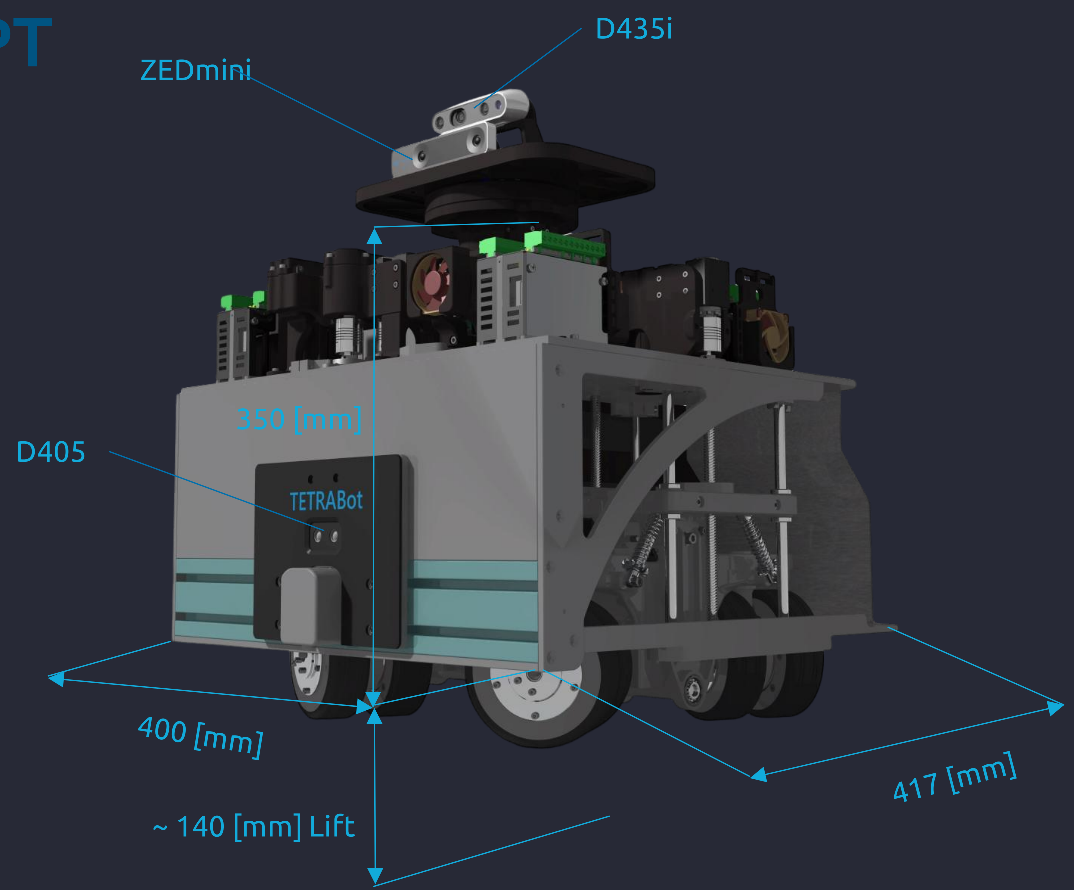
Next Steps / WIP

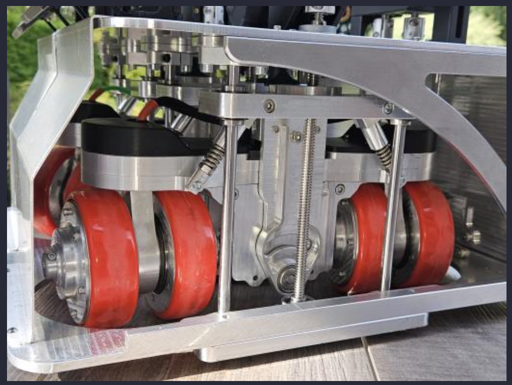
- automated docking
- automation of tool change
- SLAM (simultaneous localisation and mapping)
- Communication by 5G network

TECHNICAL FEATURES: NEW CONCEPT

Features

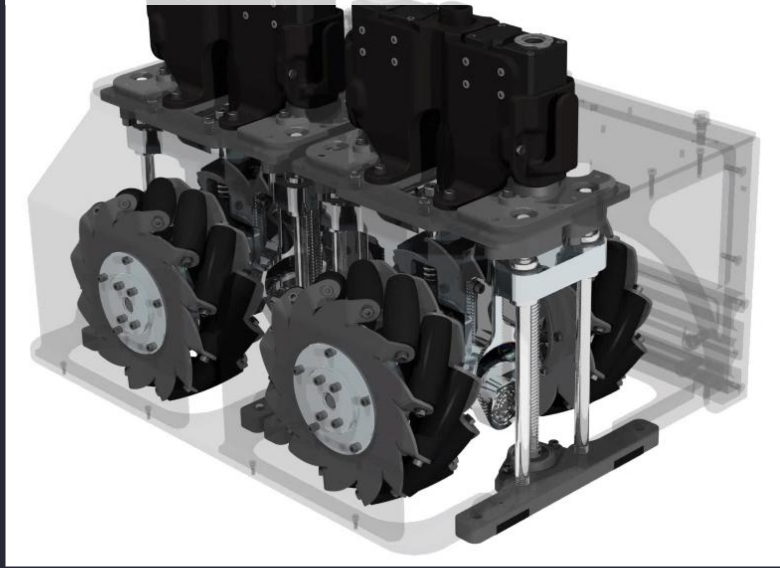
- Own weight: less 28 kg
- Size: 400 x 417 x 350 mm
- Lift load: ~ 80 kg
- Lift height: ~ 140 mm
- Regular wheels: 3DOF in the plane
- Max speed: ~ 5 km/h (depending on load and environment)
- Computer: Jetson Nano
- Operating System: Ubuntu, ROS1
- Communication: WLAN
- Sensors:
  - Aligned depth & RGB camera, Intel Realsense D435i
  - 6 DOF tracking sensor, ZEDmini
  - Docking Camera Intel D405
  - OpenCV
- max. battery runtime: ~ 2h operational (tbd)



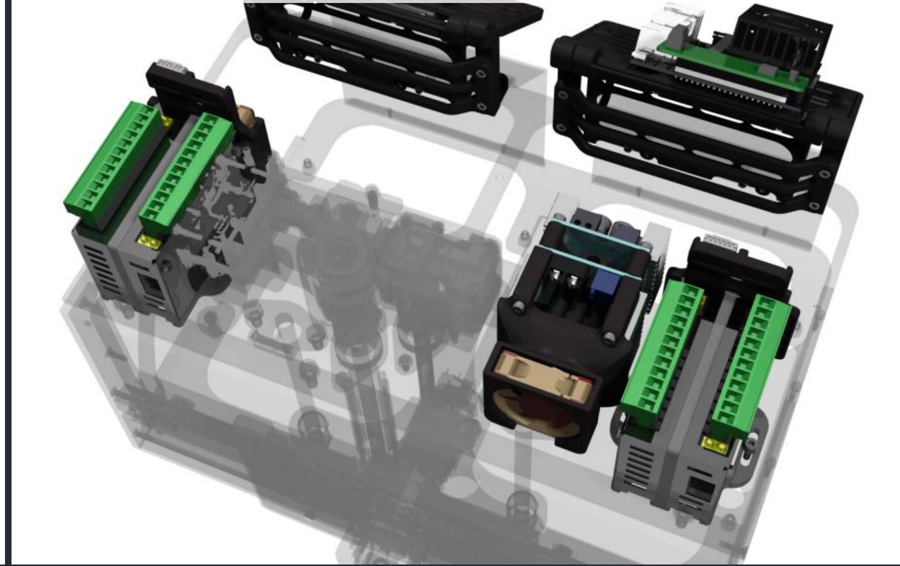


MAIN COMPONETS

Mechanic

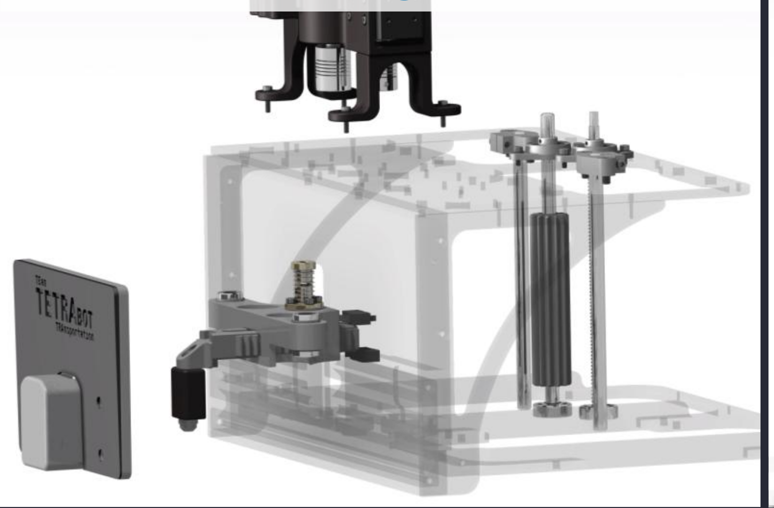


Electrics & Computer



Lift- and drive unit

Docking unit

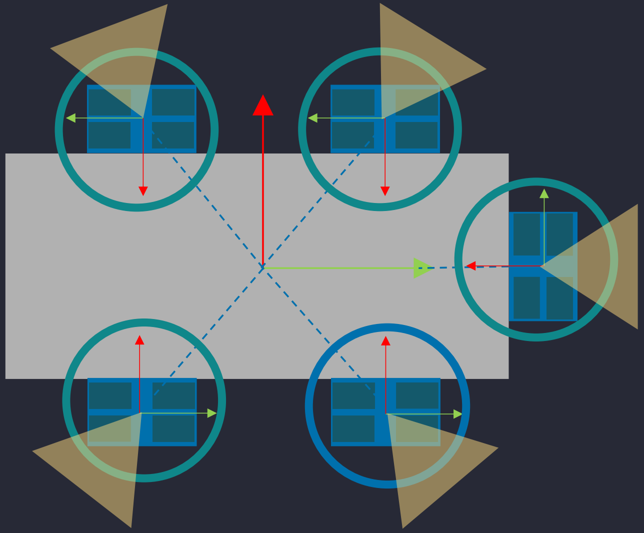


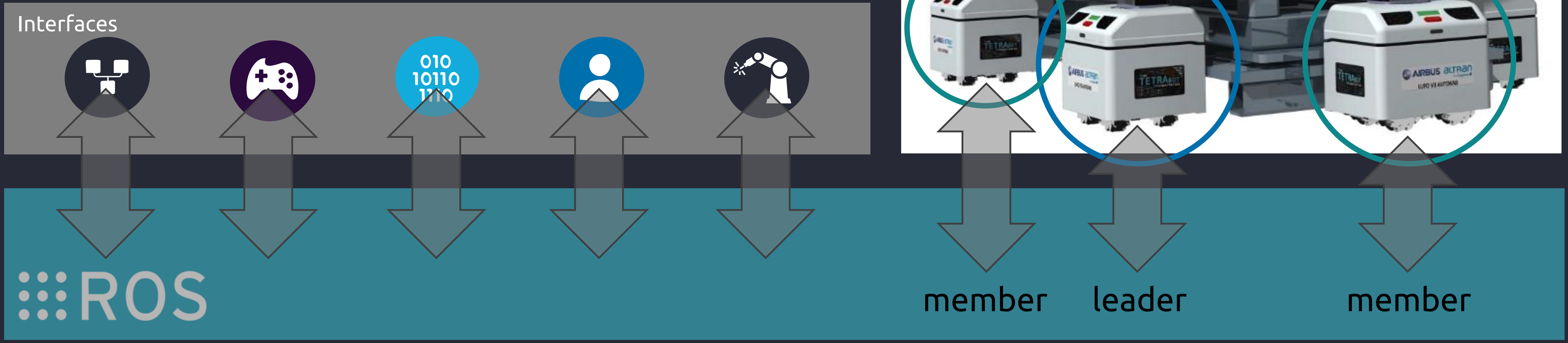
Sensorhead

TEAM COORDINATION & COMMUNICATION

Team control

- Communication by ROS
- Coordination by leader-member-concept
- Swarm behavior (e.g. share sensor data)

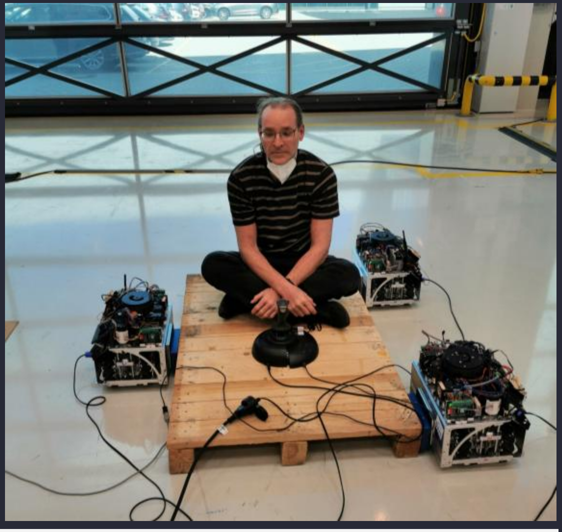




USE CASES

Use cases

- Team applications
- Single applications
- New developments

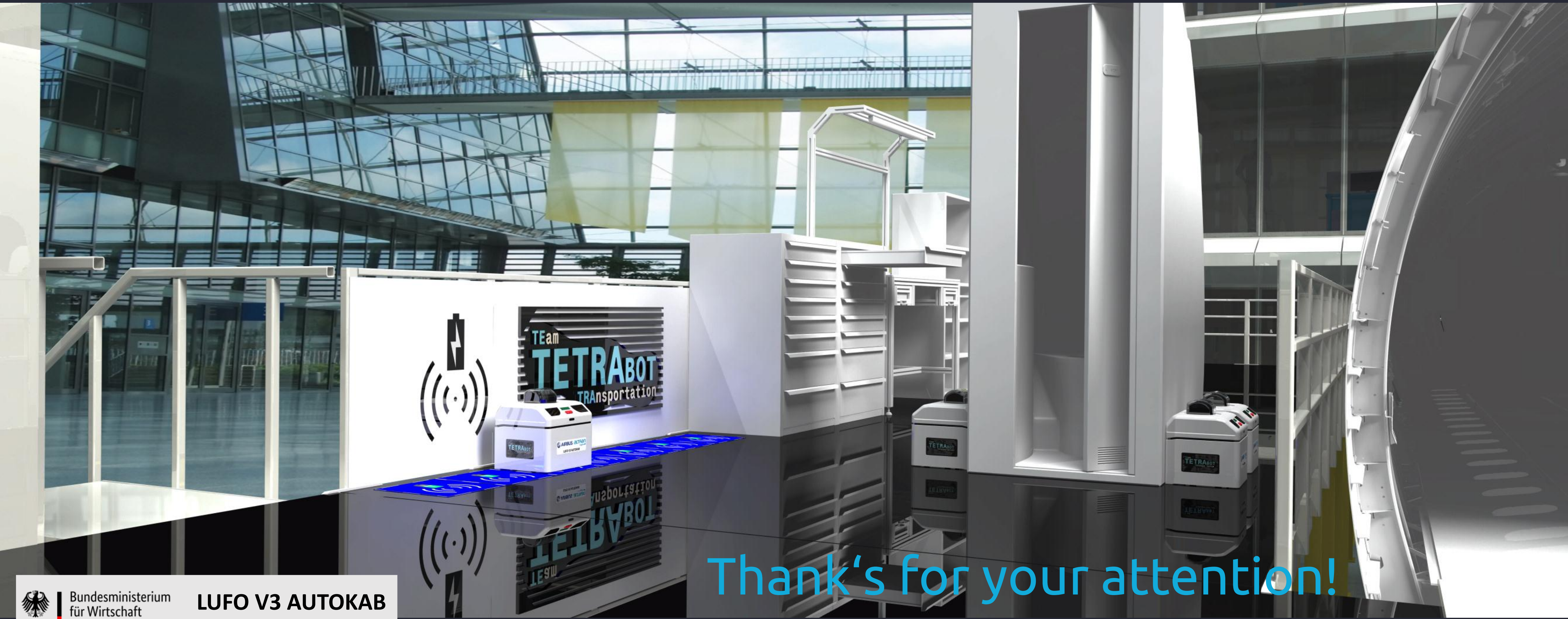


THE “TEAMS”

Thomas Stopp
Hardware

Thomas Krebs
Electrics/Software

Matthias Kossow
Software



Thank's for your attention!

Q & A

Bundesministerium für Wirtschaft und Energie

LUFO V3 AUTOKAB

Capgemini engineering

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